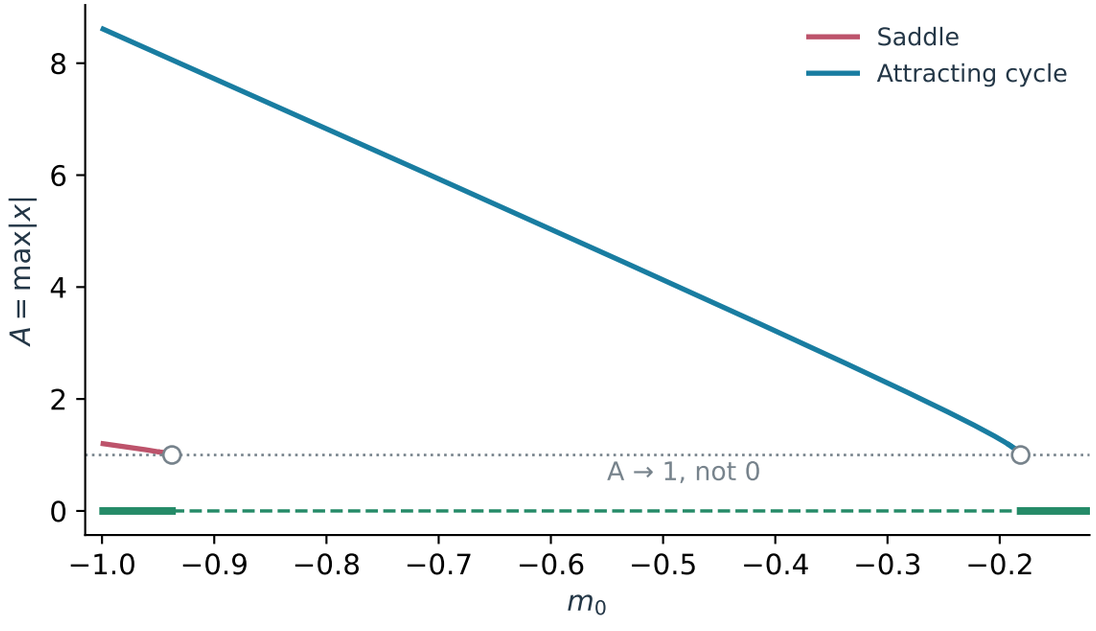
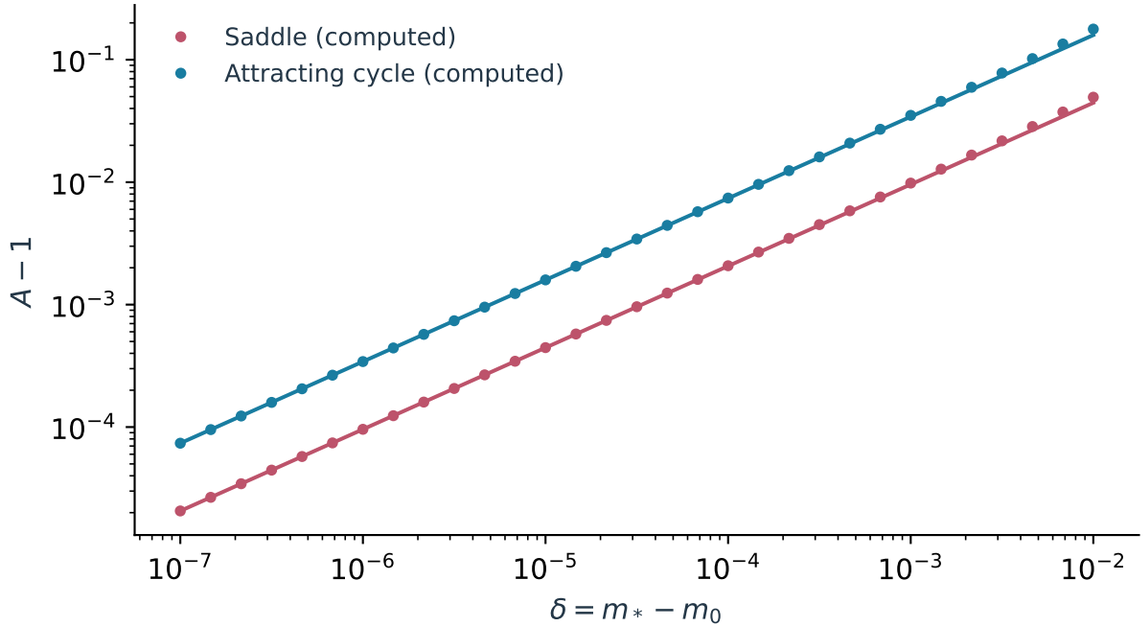
How the cycles meet the linear central region

α = 8.4, β = 12, $m_1 = -0.05$ · 31 distances per branch, from 10^{-2} to 10^{-7}

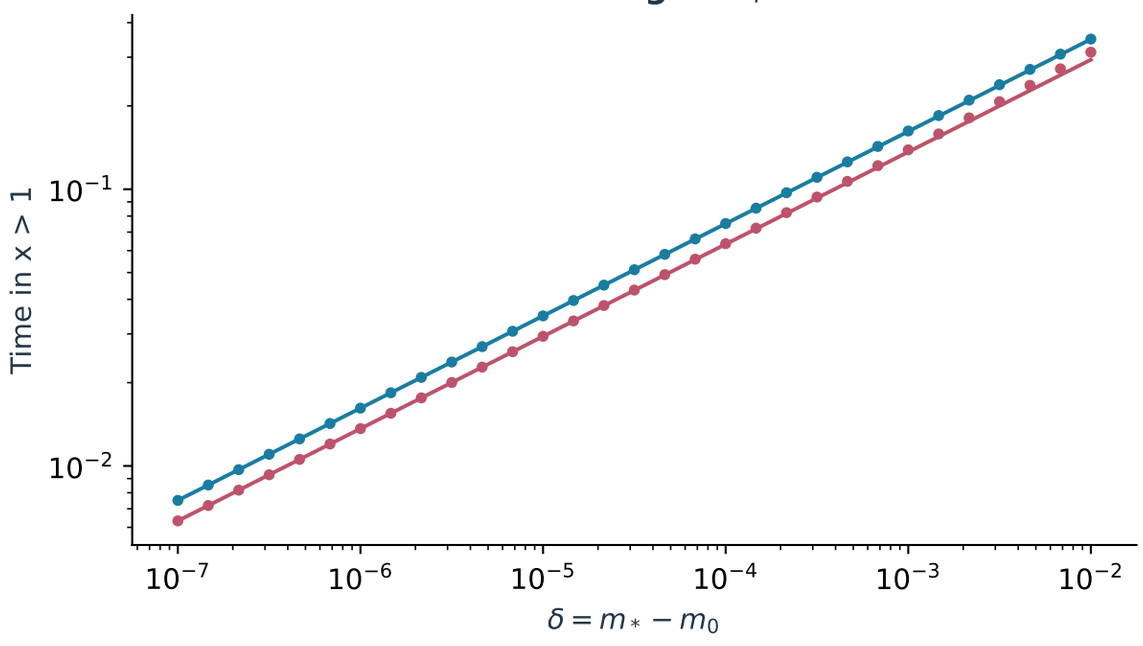
A Two distinct centre contacts



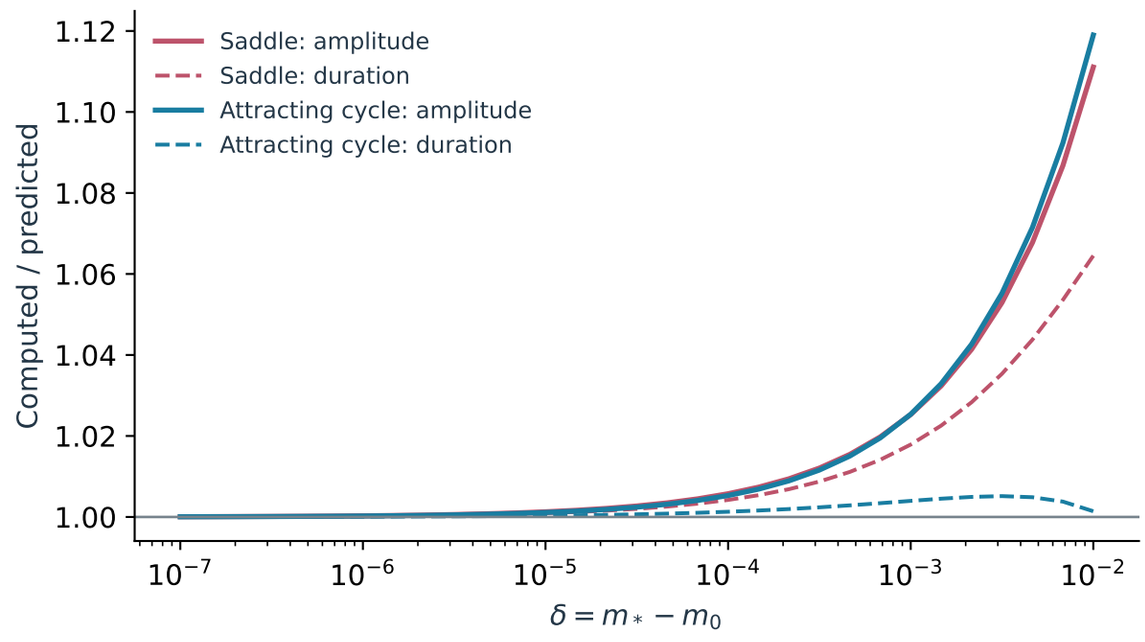
B Amplitude excess: $c_A\delta^{2/3}$



C Outer flight: $c_T\delta^{1/3}$



D Test of the coefficients



Markers: piecewise-linear closed orbits. Lines in B/C: coefficients fixed before the fine-distance sweep.
A: the green origin branch is solid where locally stable and dashed where unstable.
At δ = 10^{-7}, all four coefficient errors are below 0.0061%. This is an asymptotic numerical check, not a remainder proof.